CSE STUDY HUB

Data Structures & Algorithms

UNIT 5 · Graphs

What this unit is about

This unit builds understanding of **Graphs**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Graph terminology: vertices, edges, directed, undirected and weighted graphs

2. Graph representations: adjacency matrix and adjacency list

3. Breadth-First Search (BFS): queue-based traversal and applications

4. Depth-First Search (DFS): recursion/stack-based traversal and applications

5. BFS vs DFS; traversal order, complexity and use cases

6. Connected components, paths, cycles and spanning-tree basics

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 5 NOTES · PART 1

Graph terminology: vertices, edges, directed, undirected and weighted graphs

A graph contains vertices and edges. An adjacency matrix uses O(V²) space; an adjacency list uses O(V+E). BFS explores level by level with a queue; DFS explores deeply using recursion or a stack.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

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Connect and apply

For a practical question on Graphs, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 5 NOTES · PART 2

Breadth-First Search (BFS): queue-based traversal and applications

A queue follows FIFO (first in, first out). Enqueue inserts at the rear and dequeue removes from the front. Circular queues reuse freed array positions; deques allow operations at both ends.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Depth-First Search (DFS): recursion/stack-based traversal and applications

A stack follows LIFO (last in, first out). push inserts at the top, pop removes the top, and peek reads it. Check overflow for fixed arrays and underflow before popping an empty stack.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

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UNIT 5 NOTES · PART 3

BFS vs DFS; traversal order, complexity and use cases

For BFS vs DFS; traversal order, complexity and use cases, connect the concept to a practical scenario. Identify the input or need, the technology/process used, the output or benefit, and one limitation. Strong examples explain not only what is used but why it is suitable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connected components, paths, cycles and spanning-tree basics

A tree is a hierarchical, connected acyclic structure. Learn root, leaf, parent, child, depth and height. Traversals visit nodes in a defined order; BST ordering enables efficient search when balanced.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Graphs, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Graph terminology: vertices, edges, directed, undirected and weighted graphs** — know the definition, core working and one example.

• **Graph representations: adjacency matrix and adjacency list** — know the definition, core working and one example.

• **Breadth-First Search (BFS): queue-based traversal and applications** — know the definition, core working and one example.

• **Depth-First Search (DFS): recursion/stack-based traversal and applications** — know the definition, core working and one example.

• **BFS vs DFS; traversal order, complexity and use cases** — know the definition, core working and one example.

• **Connected components, paths, cycles and spanning-tree basics** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Graphs. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Graph terminology: vertices, edges, directed, undirected and weighted graphs and explain its importance.

Q2. Explain the working/steps involved in Graph representations: adjacency matrix and adjacency list.

Q3. Differentiate between two related concepts from Graphs.

Q4. Give one practical application of Connected components, paths, cycles and spanning-tree basics and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Graphs.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.